

Covariance section review

Reference gaps and technical questions for independent checking
Prepared 28 September 2026

Draft reviewed: Chris Addis, Constraints and biases: impact of light-cone effects on joint power spectrum and bispectrum forecasts, file **Forecast_pkbk (18).pdf**, compiled 28 September 2026. Main scope: Section 4 (pp. 6-10) and Appendix C (p. 23), with a related consistency check on p. 18. All draft equation and page numbers below refer to that version.

Assessment. The section already cites much of the central literature, including Biagetti et al. (2022), Salvalaggio et al. (2024), Rizzo et al. (2023), and Karagiannis et al. (2024). The most useful additions are Sugiyama et al. (2020) and Montano et al. (2026). The main technical concern is the distinction between the general covariance derivation and the squeezed-limit approximation used in the forecasts.

This note records a reading of the draft and a literature check. It does not independently reproduce the code or numerical forecasts. The technical concerns are questions to verify, not established implementation errors. Read it alongside the original draft, which is not embedded here.

References to add or reposition

1. Sugiyama, Saito, Beutler & Seo (2020) - high priority

[Perturbation theory approach to predict the covariance matrices of the galaxy power spectrum and bispectrum in redshift space](#). Treats joint redshift-space covariance, non-Gaussian terms, shot noise, and connected five- and six-point contributions. Cite near the correlator decomposition and at the start of Section 4.3. The existing Sugiyama et al. (2019) bibliography entry is a different paper.

2. Montano, Camera, Sefusatti & Elkhachab (2026) - high priority

[Complex yet Hermitian: Gaussian covariance of cross-correlation and multi-tracer power spectra](#). Submitted 10 June 2026. Directly relevant to complex covariance, odd parity, and multiple tracers. Cite alongside Paper II in the opening of Section 4. It explicitly recovers a result from the earlier Addis et al. paper and helps position the bispectrum extension. It is distinct from the January 2026 Montano paper already listed.

3. Chan, Moradinezhad Dizgah & Noreña (2018)

[Bispectrum Supersample Covariance](#). Discusses super-survey contributions to both bispectrum covariance and power spectrum-bispectrum cross-covariance. Cite when stating what is omitted. Distinguish coupling through measured shared modes from super-sample covariance; the latter is not established to be negligible for these forecasts merely by this citation.

4. Response formalism and historical context

[Barreira & Schmidt \(2017\), Responses in Large-Scale Structure](#), is a useful foundation for the response R^{bc} and Eqs. (39)-(40). Move or repeat the existing [Barreira \(2019\)](#) citation there. Optionally add [Sefusatti et al. \(2006\), Cosmology and the Bispectrum](#), near Eq. (19), for earlier joint P+B covariance and beat-coupling work.

Technical questions to verify

A. Gaussian symmetry factor: Eqs. (33)-(34), pp. 7-8

The single-tracer reduction to $s_B L_{\ell_1}(\mu_1) L_{\ell_2}(\mu_2)$ appears too general for higher multipoles. When equal-length sides are exchanged, the second Legendre weight can instead depend on μ_2 or μ_3 . Single-tracer symmetry of the power-spectrum product does not remove that angular dependence.

The distinction is specific: the usual factor works for scalene triangles, and for isosceles triangles with a unique weighted side $k_1 > k_2 = k_3$. It needs checking when k_1 is among the equal sides, such as $k_1 = k_2 > k_3$ or an equilateral triangle. The draft acknowledges the permutation issue immediately before Eq. (33), but the displayed formula does not show this restriction.

Literature comparison: [Salvalaggio et al. \(2024\), Eq. \(13\) in the arXiv v1 HTML](#), retains separate angular weights for the allowed permutations. Account for the conjugation convention when comparing signs. This may be a presentation issue if the implementation retains the full permutation sum.

Suggested diagnostic: in the infinitesimally thin, single-tracer equilateral case, check whether permutation symmetry makes the symmetrised dipole weight vanish because the three wavevectors sum to zero. Compare this with the positive naive weight $6L_1(\mu_1)^2$. Finite-width bins require a separate check.

B. Scope of the non-Gaussian approximation: Eqs. (40)-(41)

The transition from the general PT term to a separable response expression is the decisive approximation. The draft motivates it when the shared mode is soft, then applies the expression to all nine pairings. State this extrapolation prominently: the forecasts use a model motivated by the squeezed limit for the leading non-Gaussian terms.

The proposed wording should preserve the distinction between the general BB/PT/PB decomposition and its approximate evaluation. Keep the odd-multipole BB-PT cancellation explicitly tied to the collapsed approximation and the real-bispectrum limit used to derive it; it is not a general proof that all non-Gaussian contributions to odd multipoles are negligible.

C. Positivity of the joint P+B covariance: p. 10

Positive semidefiniteness of Eq. (41) alone does not establish positive semidefiniteness of the full block covariance. The proposed shared-response construction may guarantee it, but the connection should be demonstrated. For an invertible, positive-definite C_{PP} , the equivalent Schur-complement requirement is:

$$C_{BB} - C_{BP} C_{PP}^{-1} C_{PB} \geq 0.$$

A short construction establishing this condition would be stronger than the example minimum eigenvalues. The eigenvalue results are useful diagnostics, but apply to the tested cases. This is a request for a clearer argument, not a claim that the adopted matrix fails it.

Validation, consistency, and review brief

D. What Appendix C establishes, and what remains open

The reported agreement of 0.3% in Newtonian bispectrum signal-to-noise and 2% for the local relativistic contribution is encouraging. Those checks concern a particular single-tracer Euclid-like bin and compare the collapsed approximation with a tree-level exchange-only calculation. They do not directly establish the accuracy of marginalised f_{NL} errors or of multi-tracer forecasts.

Useful further checks would be the change in at least one headline parameter constraint under a justified covariance variation, and sensitivity to excluding flattened bins where the mode-counting comparison differs by approximately 25%. The comparison with T_{2211} should not be described as validation against the complete tree-level trispectrum, because T_{3111} is absent.

The attribution of negative eigenvalues in the exchange-only calculation to missing Z_3 / T_{3111} terms is plausible, but should remain a possible explanation unless restoring those terms is shown to resolve the issue. Positive semidefiniteness is a consistency condition; it does not by itself demonstrate physical accuracy.

Two direct consistency fixes

Draft p. 18: the limitations discussion still says that non-Gaussian covariance and C_{PB} are neglected. This conflicts with Section 4.3. Reconcile the wording with the covariance actually used for each reported result.

Bibliography: update Salvalaggio et al. from the arXiv-only entry to *JCAP* **08** (2024) 046, [doi:10.1088/1475-7516/2024/08/046](https://doi.org/10.1088/1475-7516/2024/08/046).

Suggested questions for an independent reviewer or LLM

Assess these points against the original draft and primary sources, rather than assuming the review is correct. For each point, distinguish a confirmed error, a valid approximation needing clearer wording, and a concern that is unsupported or depends on the estimator convention.

1. Re-derive the allowed Wick permutations behind Eqs. (33)-(34), including their Legendre weights. Identify exactly which triangle shapes and multipoles permit the stated symmetry-factor reduction.
2. Check the assumptions behind Eqs. (40)-(41), their use for all side pairings, and the scope of the odd-multipole cancellation. Does the stated shot-noise treatment follow from the same response model?
3. Determine whether the proposed shared-response model establishes the joint Schur-complement condition, including C_{PB} and the Gaussian C_{PP} .
4. Assess whether Appendix C supports the claimed forecast accuracy, and whether the omitted references materially change how the contribution should be positioned.

Additional primary sources already relevant to the draft

[Biagetti et al. \(2022\), The Covariance of Squeezed Bispectrum Configurations](#)

[Rizzo et al. \(2023\), The Halo Bispectrum Multipoles in Redshift Space](#)

[Salvalaggio et al. \(2024\), Bispectrum non-Gaussian Covariance in Redshift Space](#)